

UNIT -1

: Introduction Concepts Introduction to networks – Data Communication – Data flow simplex, Half duplex, Full Duplex-Type of Connection – broadcast, Point-to-Point, multi-drop. Bandwidth- bit rate, baud rate. Transmission media –Copper wires, fibre optics, Radio transmission, microwave, Satellite. Switching - circuit, packet, message, ISDN.

Computer Networks – Unit I Notes (Short & Easy Language)

Unit-I: Introduction Concepts

1. Introduction to Networks

A **computer network** is a group of two or more computers or devices connected together to share information and resources.

Purpose of a Network

- Share data and files
- Share printers and internet
- Communication between users
- Resource sharing
- Easy access to information

Examples

- College computer lab
- Office network
- Internet
- Wi-Fi at home

2. Data Communication

Data Communication means the exchange of data between two or more devices through a communication medium.

Components of Data Communication

1. Sender
2. Receiver
3. Message (Data)
4. Communication Medium (Cable/Wireless)
5. Protocol (Rules of communication)

Example

A student sends an email to a teacher using the Internet.

3. Data Flow Modes

Data flow means the direction in which data travels between two devices.

A. Simplex Mode

- Data flows in **one direction only**.
- One device sends, the other only receives.

Example

- Keyboard → Computer
- Television Broadcast

Advantages

- Simple
- Low cost

Disadvantages

- No reply is possible.

B. Half-Duplex Mode

- Data flows in **both directions**, but **one direction at a time**.

Example

- Walkie-Talkie

Advantages

- Two-way communication
- Less expensive

Disadvantages

- Devices cannot send and receive simultaneously.

C. Full-Duplex Mode

- Data flows in **both directions at the same time**.

Example

- Mobile Phone
- Video Call

Advantages

- Fast communication
- No waiting time

Disadvantages

- More expensive

Comparison of Data Flow Modes

Mode	Direction	Simultaneous Communication	Example
Simplex	One-way	No	Keyboard, TV
Half-Duplex	Two-way	No	Walkie-Talkie
Full-Duplex	Two-way	Yes	Mobile Phone

4. Types of Connection

A connection is the method used to connect devices in a network.

A. Broadcast Connection

- One sender sends data to **all devices** on the network.

- Every device receives the message.

Example

- Television broadcasting
- Radio broadcasting

Advantages

- Easy to send information to everyone.

B. Point-to-Point Connection

- Data is sent between **only two devices**.

Example

- Computer to Printer
- Mobile to Mobile (Bluetooth)

Advantages

- Secure
- Fast communication

C. Multi-Drop Connection

- One communication line is shared by **multiple devices**.

Example

- Several computers connected to one cable.

Advantages

- Low cost
- Easy installation

5. Bandwidth

Bandwidth is the **maximum amount of data** that can be transmitted over a network in one second.

Unit

- Bits per second (bps)

Example

A connection with 100 Mbps bandwidth can transfer up to 100 million bits every second.

Higher Bandwidth

- Faster data transfer
- Better internet speed

6. Bit Rate

Bit Rate is the **number of bits transmitted in one second**.

Unit

- bps (bits per second)

Example

100 Mbps means 100 million bits are transferred every second.

7. Baud Rate

Baud Rate is the **number of signal changes (symbols) transmitted per second**.

Unit

- Baud

Difference

Bit Rate	Baud Rate
Number of bits/sec	Number of signals/sec
Unit: bps	Unit: Baud

8. Transmission Media

Transmission media is the **path through which data travels**.

There are two main types:

- Guided (Wired)
- Unguided (Wireless)

A. Copper Wire

Copper cables carry electrical signals.

Examples

- Twisted Pair Cable
- Coaxial Cable

Advantages

- Low cost
- Easy to install

Disadvantages

- Limited distance
- Signal loss

B. Fibre Optic Cable

Fibre optic cable carries data using **light signals**.

Advantages

- Very high speed
- Long distance
- High security
- No electromagnetic interference

Disadvantages

- Expensive
- Difficult to install

C. Radio Transmission

Radio waves transmit data without wires.

Examples

- FM Radio
- Wi-Fi
- Bluetooth

Advantages

- Wireless communication

- Easy installation

Disadvantages

- Interference may occur.

D. Microwave Transmission

Microwaves transmit data between two towers.

Advantages

- High-speed communication
- Long-distance transmission

Disadvantages

- Requires line-of-sight (clear path).

E. Satellite Communication

Satellites receive and send signals over very long distances.

Uses

- TV Broadcasting
- GPS
- Internet
- Weather Forecasting

Advantages

- Covers large areas
- Useful in remote places

Disadvantages

- High cost
- Signal delay

9. Switching

Switching is the process of moving data from the sender to the receiver through a network.

A. Circuit Switching

- A **dedicated communication path** is established before communication starts.
- The path remains reserved until communication ends.

Example

Traditional Telephone Network

Advantages

- Reliable communication
- No data loss during transmission

Disadvantages

- Wastes bandwidth if the line is idle.

B. Packet Switching

- Data is divided into **small packets**.

- Each packet travels independently.

- Packets are reassembled at the destination.

Example

Internet

Advantages

- Efficient use of bandwidth
- Fast communication

Disadvantages

- Packets may arrive out of order.

C. Message Switching

- The complete message is sent from one node to another.
- Each node stores the message before forwarding it.

Advantages

- No dedicated path required

Disadvantages

- Slow due to storage delay

D. ISDN (Integrated Services Digital Network)

ISDN is a **digital communication technology** that provides voice, video, and data services over telephone lines.

Features

- Digital transmission
- Better quality than analog
- Supports voice, video, and data together

Applications

- Video conferencing
- Internet access
- Telephone communication

Comparison of Switching Techniques

Switching	Path Required	Speed	Example
Circuit Switching	Dedicated Path	Medium	Telephone
Packet Switching	No Dedicated Path	Fast	Internet
Message Switching	Store and Forward	Slow	Postal-like Messaging
ISDN	Digital Network	Fast	Digital Telephone

Key Points for Exam (Remember)

- Computer Network:** Connection of two or more devices for sharing resources.
- Data Communication:** Exchange of data between devices.
- Simplex:** One-way communication.
- Half-Duplex:** Two-way, one direction at a time.
- Full-Duplex:** Two-way, both directions simultaneously.

- **Broadcast:** One sender to all devices.
- **Point-to-Point:** One sender to one receiver.
- **Multi-Drop:** One communication line shared by multiple devices.
- **Bandwidth:** Maximum data transfer capacity.
- **Bit Rate:** Number of bits transmitted per second.
- **Baud Rate:** Number of signal changes per second.
- **Copper Wire:** Low cost but limited distance.
- **Fibre Optics:** High speed, long distance.
- **Radio:** Wireless communication.
- **Microwave:** High-speed, line-of-sight communication.
- **Satellite:** Covers very large geographical areas.
- **Circuit Switching:** Dedicated communication path.
- **Packet Switching:** Data divided into packets (used by the Internet).
- **Message Switching:** Store-and-forward transmission.
- **ISDN:** Digital network supporting voice, video, and data.

Exam Tip: Learn the definitions, advantages, disadvantages, and examples of **Simplex**, **Half-Duplex**, **Full-Duplex**, **Transmission Media**, and **Switching Techniques**, as these are the most frequently asked topics in Unit I.

EXPLAIN IN DETAILS ALL UNIT TOPICS.....

Introduction to Computer Networks

A computer network is a group of two or more computers that are connected together to share information, resources, and services.

☞ Why do we need computer networks?

- To share files (documents, videos, etc.)
- To use shared devices (like printers, scanners)
- To communicate through emails, chats, or video calls
- To access the internet

💻 Types of Computer Networks:

Type	Description	Example
LAN (Local Area Network)	Small area network	Office, School
MAN (Metropolitan Area Network)	Covers a city	College campus

WAN (Wide Area Network)	Large area,	Internet, country or world
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🏢 2. Network Models

Network models are like frameworks that show how computers talk to each other in a network. They help in designing and understanding networks.

🧱 3. OSI Model (Open Systems Interconnection)

The **OSI (Open Systems Interconnection)** model is a conceptual framework that standardizes how different computer systems communicate over a network. It divides the communication process into seven layers, each with specific functions. This layered approach allows for greater flexibility, interoperability, and easier troubleshooting of network issues.

Here's a breakdown of the seven layers:

1. **Physical Layer:** This layer deals with the physical connection between devices, including the cables, network interface cards, and the transmission of raw data bits (0s and 1s). It defines the physical characteristics of the network, such as voltage levels, data rates, and the physical topology.
2. **Data Link Layer:** Responsible for reliable data transfer between two directly connected nodes (devices on the same network). It packages data into frames, handles error detection and correction, and manages access to the physical medium (e.g., using MAC addresses).
3. **Network Layer:** Handles routing data packets between different networks. It uses IP addresses to identify devices and determines the best path

for data to travel from the source to the destination.

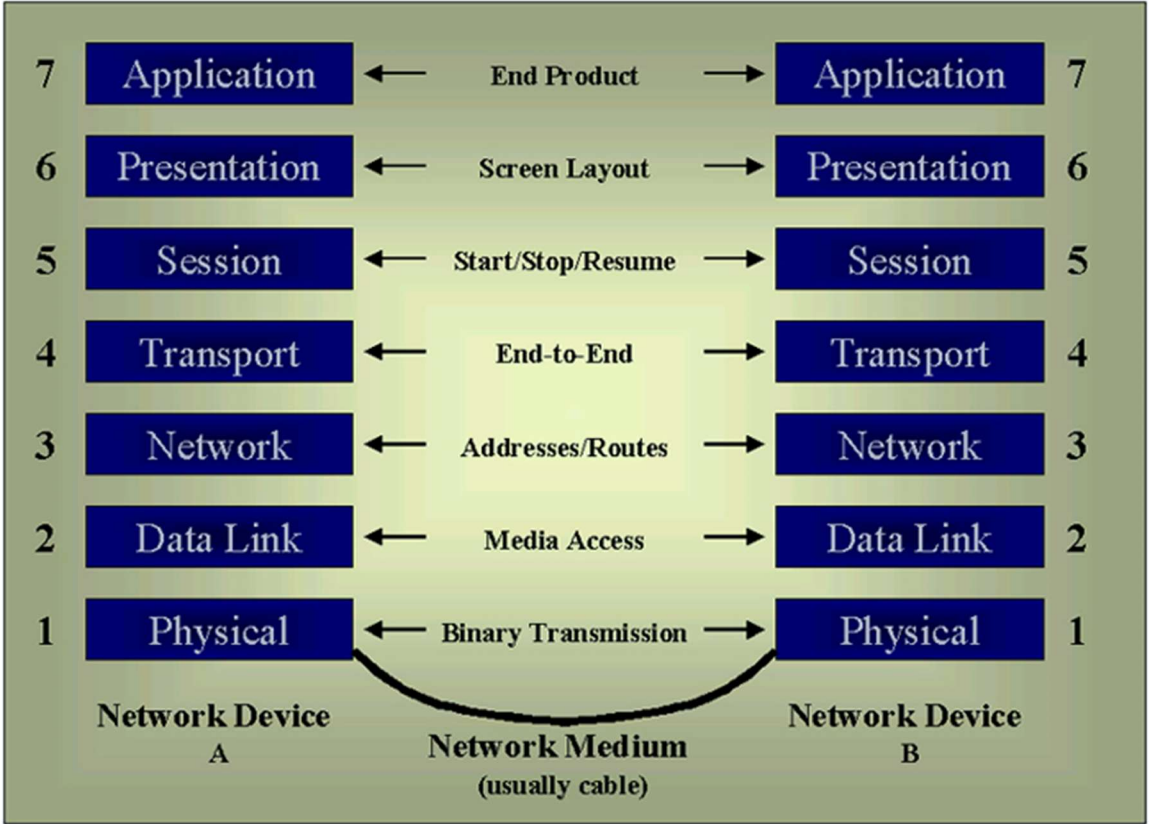
4. **Transport Layer:** Provides reliable, end-to-end communication between applications. It manages data flow control, error checking, and segmentation/reassembly of data into manageable packets (or segments).

5. **Session Layer:** Establishes, manages, and terminates connections (sessions) between applications on different devices. It also handles authentication and reconnection.

6. **Presentation Layer:** Focuses on data formatting, encryption, and compression. It ensures that data is in a format that the receiving application can understand.

7. **Application Layer:** This is the layer closest to the user, providing network services to applications. It includes protocols for things like file transfer (FTP), email (SMTP), and web browsing (HTTP).

In essence, the OSI model provides a structure for understanding how data flows through a network, from the physical transmission of bits to the user-facing application layer.



The OSI model has 7 layers, and each layer has a specific job in the communication process. It was created by ISO (International Organization for Standardization).

📁 OSI Model Layers (Bottom to Top):

Layer No.	Layer Name	Simple Function
1	Physical Layer	Sends raw bits (0s and 1s) over cables
2	Data Link Layer	Ensures error-free data between connected devices
3	Network Layer	Decides best path for data (IP addressing)
4	Transport Layer	Ensures complete and correct data delivery (TCP/UDP)
5	Session Layer	Manages communication sessions
6	Presentation Layer	Converts data into readable format (like encryption)
7	Application Layer	User interaction layer (browser, email, etc.)

🔄 Easy Example:

If you're sending a WhatsApp message:

Application Layer: WhatsApp app
Transport Layer: Ensures message is delivered correctly
Network Layer: Finds the path to the other phone
Physical Layer: Sends data through Wi-Fi/mobile data

🌐 4. TCP/IP Model (Transmission Control Protocol / Internet Protocol)

This is a practical and real-world model used for internet communication. It has 4 layers instead of 7.

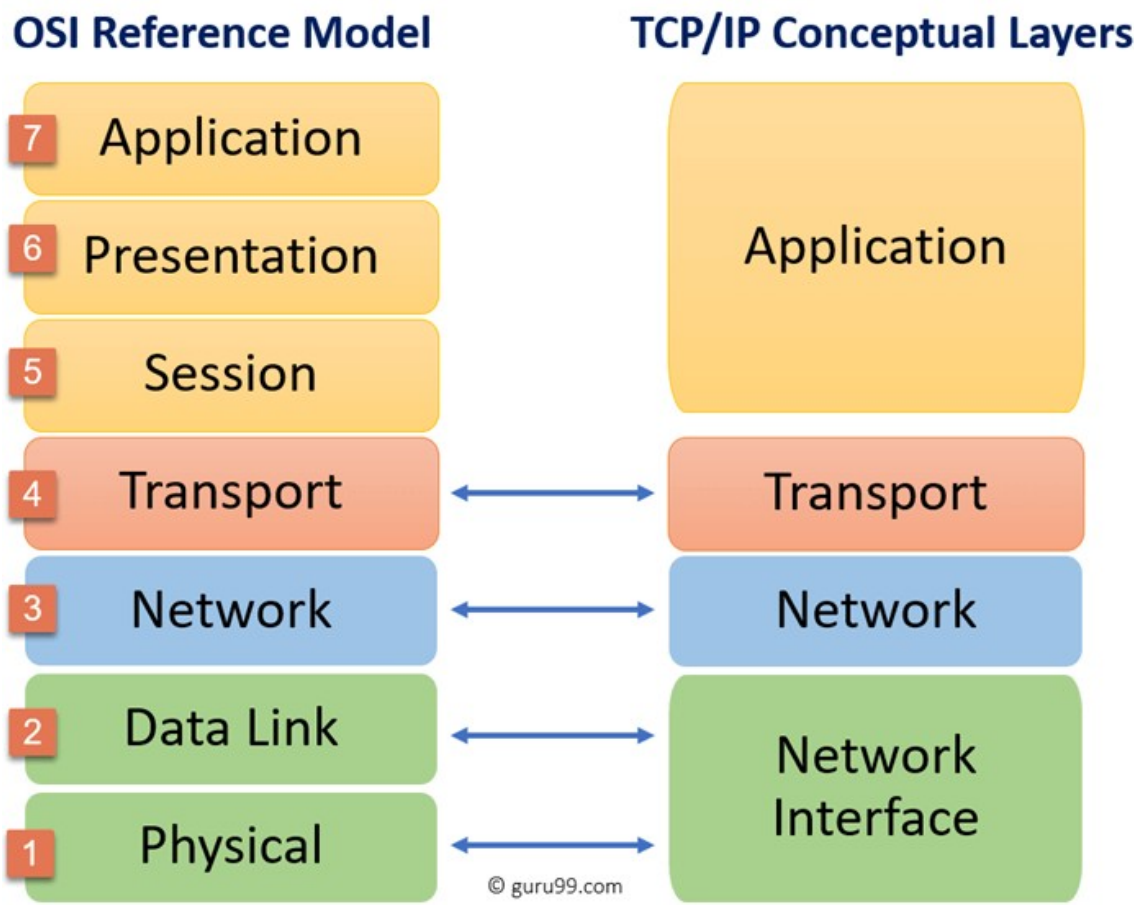
📁 TCP/IP Model Layers:

Layer No.	Layer Name	Function
1	Network Access Layer	Deals with physical delivery (LAN, Wi-Fi)
2	Internet Layer	Sends data from one device to another using IP
3	Transport Layer	Ensures reliable delivery using TCP/UDP
4	Application Layer	Where applications (web, email) work

📊 OSI vs TCP/IP

OSI Model	TCP/IP Model
Has 7 layers	Has 4 layers
More theoretical	Practical and used on internet
Developed by ISO	Developed by the U.S. Dept. of Defense

📊 Diagram: OSI vs TCP/IP Model (Simple View)



UNIT 1 END

ASSISGNMENT QUESTION
UNIT-1

◆ Introduction to Computer Networks

1. Define a computer network. Explain its main objectives.
2. What are the key advantages and disadvantages of computer networking?
3. Explain the difference between LAN, MAN, and WAN with examples.
4. Compare Peer-to-Peer and Client-Server networks with diagrams.
5. What are different types of network topologies? Explain with diagrams.

ASSIGNMENT UNIT1

Q1. Explain the fundamentals of computer networks and discuss the importance of data communication in modern networking. (Descriptive Assignment)

Q2. A communication channel has a bandwidth of 5 MHz and transmits data at 20 Mbps. Calculate the spectral efficiency (bits/s/Hz) of the channel. (Descriptive Assignment)

Q3. A university plans to connect multiple academic buildings using a high-speed communication network. Analyze the available transmission media and recommend the most suitable option with proper justification.

(Case Study Assignments)

Q4. Explain how full-duplex communication improves the performance of video conferencing and online meetings.

(Application-Based Assignments)

Q5. Prepare a short literature review on recent developments in high-speed transmission media used in modern computer networks. (Research-Oriented Assignments)

QUIZ UNIT-1

Q.No.	Question	Options	Answer	Marks
Q1	Which communication mode allows data transmission in only one direction?	a) Full Duplex b) Half Duplex c) Simplex d) Multiplex		1
Q2	Which connection type delivers data to all devices on a network?	a) Point-to-Point b) Broadcast c) Multi-drop d) ISDN		1
Q3	Which parameter measures the number of bits transmitted per second?	a) Baud Rate b) Bit Rate c) Bandwidth Delay d) Frequency		1
Q4	Which transmission medium provides the highest bandwidth?	a) Copper Wire b) Microwave c) Fibre Optics d) Radio Waves		1
Q5	Which switching technique establishes a dedicated communication path before transmission?	a) Packet Switching b) Message Switching c) Circuit Switching d) Broadcast Switching		1

SURPRISE CLASS TEST UNIT-1

Q.No.	Question Type	Question	Marks
Q1	Descriptive	Explain the concept of a computer network .	0.5
Q2	Descriptive	Differentiate between Simplex and Full-Duplex communication.	0.5
Q3	Numerical / Problem-Solving	A modem transmits data at 9600 bps with a baud rate of 2400 . Calculate the number of bits per signal element .	0.5
Q4	Numerical / Problem-Solving	A communication channel has a bandwidth of 5 MHz and a bit rate of 20 Mbps . Calculate the spectral efficiency (bits/s/Hz) .	0.5
Q5	Case Study	A school wants to connect two nearby buildings. Recommend the most suitable transmission medium with a reason.	0.5
Q6	Case Study	A television broadcasting company needs to send the same signal to thousands of users. Which type of connection is most suitable? Justify your answer.	0.5
Q7	Application-Based	Explain how packet switching is used in Internet communication.	0.5
Q8	Application-Based	Suggest the most suitable data communication mode for an online video conference and justify your choice.	0.5
Q9	Research-Oriented	List any two recent applications of fibre optic communication in modern networks.	0.5
Q10	Research-Oriented	Compare Circuit Switching and Packet Switching based on any two parameters .	0.5

END.....ASSISGNMENT QUESTION UNIT-1

MINOR CLASS TEST UNIT -1